

$X \rightarrow VV$ angular analysis

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Motivation

- At least two reaction in LHCb demanding the amplitude analysis of $X \rightarrow VV$
- The problem looks like fun
- Biplab keeps sending reminders of the seminar :)

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Setup: $X \rightarrow VV$

- X is a particle produced in strong interaction prompt pp collision
- V is a vector particle, consider $J/\psi, \phi, Z^0, \gamma$
- VV are identical

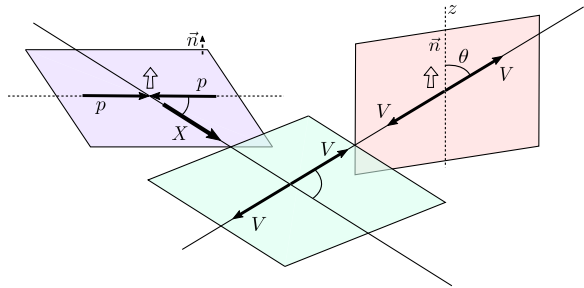
$X \rightarrow VV$ amplitude

Amplitude:

$$A_{\lambda_1, \lambda_2}^M = n_J d_{M, \lambda_1 - \lambda_2}^J(\theta) H_{\lambda_1, \lambda_2}(-1)^{1 - \lambda_2}$$

Differential width (intensity):

$$I(\theta) = \sum_M \rho_M \sum_{\lambda_1, \lambda_2} |A_{\lambda_1, \lambda_2}^M|.$$



Production by pp collision

- parity conservation $\Rightarrow$ no longitudinal polarisation
- moreover, can only be polarized perpendicular to the production plane $\vec{n}$
- choose $z \uparrow \vec{n} \Rightarrow$ diagonal polarization matrix $\delta_{M, M'} \rho_M$

Intensity for the unpolarized decay

$$I(\theta) = n_J^2 \sum_{M, \lambda_1, \lambda_2} \rho_M |d_{M, \lambda_1 - \lambda_2}^J(\theta)|^2 |H_{\lambda_1, \lambda_2}|^2$$

What is no polarization $\rho_M = 1$ (quite likely in pp)?

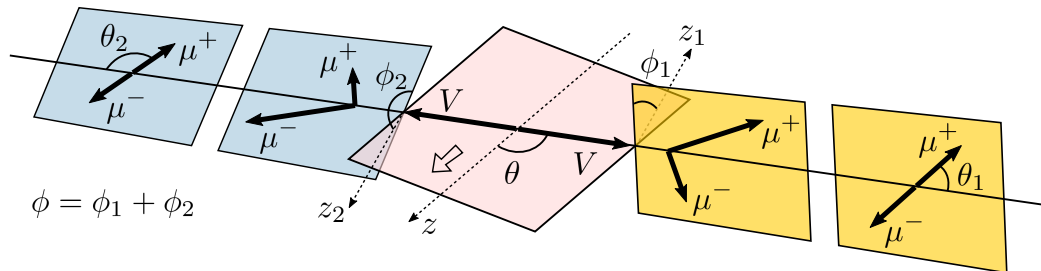
$$I(\theta) = n_J^2 \sum_{\lambda_1, \lambda_2} |H_{\lambda_1, \lambda_2}|^2$$

no J : explicit dependence on J disappears. What helps to determine J ?

- (a) Can helicity couplings tell us something?
- (b) Will the decays of V help?

Adding vector decays

Four-body decay angles



- θ is the polar angle of $(J/\psi)_1$ with respect to the polarization direction
- (θ_1, ϕ_1) are the spherical angles of μ^+ in the $(J/\psi)_1$ helicity frame
- (θ_2, ϕ_2) are the spherical angles of μ^+ in the $(J/\psi)_2$ helicity frame

No polarization $\Rightarrow$ no $z \Rightarrow$ no decay plane (pink) $\Rightarrow$ only $\phi = \phi_1 + \phi_2$ matters.

$X \rightarrow V(\mu, \mu) V(\mu, \mu)$ amplitude

$$\begin{aligned}
 A_{\xi_1, \xi_2}^M &= n_J \sum_{\lambda_1, \lambda_2} d_{M, \lambda_1 - \lambda_2}^{\textcolor{red}{J}}(\theta) H_{\lambda_1, \lambda_2} (-1)^{1 - \lambda_2} \\
 &\quad \times n_1 D_{\lambda_1, \xi_1}^{1*}(\phi_1, \theta_1, 0) \\
 &\quad \times n_1 D_{\lambda_2, \xi_2}^{1*}(\phi_2, \theta_2, 0)
 \end{aligned}$$

- $\xi_i = \lambda_{\mu^+, i} - \lambda_{\mu^-, i}$ difference of muon helicities
- $\xi \in \{-1, 1\}$ since $\xi = 0$ is suppressed by $m_\mu/m_{J/\psi}$.

$$F^{(\mu\mu)} = \sum_{\xi \in \{-1, 1\}} \int d \cos \theta d_{\lambda, \xi}^1(\theta) d_{\lambda', \xi}^1(\theta) = \frac{1}{3} \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{pmatrix}_{\lambda, \lambda'}$$

$X \rightarrow V(K^+, K^-) V(K^+, K^-)$ amplitude

$$\begin{aligned} A^M &= n_J \sum_{\lambda_1, \lambda_2} d_{M, \lambda_1 - \lambda_2}^{\textcolor{red}{J}}(\theta) H_{\lambda_1, \lambda_2}(-1)^{1 - \lambda_2} \\ &\times n_1 D_{\lambda_2, 0}^{1*}(\phi_1, \theta_1, 0) \\ &\times n_1 D_{\lambda_2, 0}^{1*}(\phi_2, \theta_2, 0) \end{aligned}$$

- $\phi \rightarrow K^+ K^-$ P -wave

The difference in marginal: no final state spin.

$$F^{(KK)} = \int d\cos\theta d_{\lambda, 0}^1(\theta) d_{\lambda', 0}^1(\theta) = \frac{1}{3} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}_{\lambda, \lambda'}$$

Symmetry constraints

Helicity couplings

[Martin, Spearman (1970) book]

Reappearance of J

$$|\text{two part. state}; \lambda_1, \lambda_2\rangle = |\vec{p}, \lambda_1\rangle \otimes |-\vec{p}, \lambda_2\rangle (-1)^{j_2 - \lambda_2}$$

$$|JM; \lambda_1, \lambda_2\rangle = n_J \int \frac{d\Omega}{4\pi} D_{M\lambda}^{J*} |\text{two part. state}; \lambda_1, \lambda_2\rangle$$

$$H_{\lambda_1, \lambda_2} = \langle JM; \lambda_1, \lambda_2 | \hat{T} | JM \rangle$$

Parity

$$\mathcal{P} |JM; \lambda_1, \lambda_2\rangle = (-1)^J \eta_V^2 |JM; -\lambda_1, -\lambda_2\rangle.$$

Permutation $1 \leftrightarrow 2$

$$\mathcal{P}_{12} |JM; \lambda_1, \lambda_2\rangle = (-1)^J |JM; \lambda_2, \lambda_1\rangle.$$

$\Rightarrow$

$$H_{\lambda_1, \lambda_2} = (-1)^J \eta_X H_{-\lambda_1, -\lambda_2},$$

$$H_{\lambda_1, \lambda_2} = (-1)^J H_{\lambda_2, \lambda_1}.$$

Matrix of helicity couplings

Matrix of helicity couplings

$$H_{\lambda_1, \lambda_2} = \begin{pmatrix} h_{1,1} & h_{1,0} & h_{1,-1} \\ h_{0,1} & h_{0,0} & h_{0,-1} \\ h_{-1,1} & h_{-1,0} & h_{-1,-1} \end{pmatrix}$$

The same-color elements are connected by symmetries.

The symmetry relates the couplings

$$H_{\lambda_1, \lambda_2} = (-1)^J \eta_X H_{-\lambda_1, -\lambda_2},$$

$$H_{\lambda_1, \lambda_2} = (-1)^J H_{\lambda_2, \lambda_1}.$$

Four categories of possible helicity matrices:

group	$\eta_X(-1)^J, (-1)^J$	J^P	symmetry
<i>I</i>	$+, +$	$0^+, 2^+, 4^+, 6^+$	symmetric, <i>S</i>
<i>II</i>	$-, +$	$0^-, 2^-, 4^-, 6^-$	symmetric, <i>S</i>
<i>III</i>	$+, -$	$1^-, 3^-, 5^-, 7^-$	antisymmetric, <i>A</i>
<i>IV</i>	$-, -$	$1^+, 3^+, 5^+, 7^+$	antisymmetric, <i>A</i>

$$\begin{array}{cccc}
 H^{(I)} & H^{(II)} & H^{(III)} & H^{(IV)} \\
 \begin{pmatrix} b & a & c \\ a & d & a \\ c & a & b \end{pmatrix}_S & \begin{pmatrix} b & a & \\ a & & -a \\ & -a & -b \end{pmatrix}_S & \begin{pmatrix} & a & \\ -a & & -a \\ & a & \end{pmatrix}_A & \begin{pmatrix} & a & c \\ -a & & a \\ -c & -a & \end{pmatrix}_A
 \end{array}$$

a, b, c, d are still unknown coefficients, complex in general.

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$$\begin{matrix} \boxed{0^+} \\ \begin{pmatrix} b & & \\ & d & \\ & & b \end{pmatrix}_S \end{matrix} \quad \begin{matrix} \boxed{0^-} \\ \begin{pmatrix} b & & \\ & & \\ & & -b \end{pmatrix}_S \end{matrix}$$

$$\begin{matrix} \boxed{1^+} \\ \begin{pmatrix} & a & \\ -a & & a \\ & -a & \end{pmatrix}_A \end{matrix}$$

a, b, c, d are still unknown coefficients, complex in general.

Landau-Yang theorem

"A massive particle with spin 1 cannot decay into two photons", wikipedia

Photons do not carry the longitudinal polarizarion $\Rightarrow H_{0,i} = H_{i,0} = 0$

$H^{(I)}$	$H^{(II)}$	$H^{(III)}$	$H^{(IV)}$
$\begin{pmatrix} b & a & c \\ a & d & a \\ c & a & b \end{pmatrix}_S$	$\begin{pmatrix} b & a & \\ a & & -a \\ & -a & -b \end{pmatrix}_S$	$\begin{pmatrix} & a & \\ -a & & -a \\ & a & \end{pmatrix}_A$	$\begin{pmatrix} & a & c \\ -a & & a \\ -c & -a & \end{pmatrix}_A$
$\begin{pmatrix} b & & \\ & d & \\ & & b \end{pmatrix}_S$	$\begin{pmatrix} b & & \\ & & \\ & & -b \end{pmatrix}_S$		$\begin{pmatrix} & a & \\ -a & & a \\ & -a & \end{pmatrix}_A$
0^+	0^-		1^+

No decay to two photons for 1^+ , and group-III: $1^-, 3^-, 5^-, \dots$

Hypothesis testing

Determination of the spin-parity

LLH comparison

- Only group can be determined without polarization
- Already something(!) - to distinguish $0^\pm$, $1^\pm$, $2^\pm$.

Likelihood of hypothesis (model $\mathcal{M}$):

$$\text{LLH}_{\mathcal{M}} = - \sum_{e=1}^{N_{\text{ev}}} \log I(\tau_e | \mathcal{M}\{\hat{h}\}),$$

Test statistics :

$$\text{TS}_{\mathcal{M}, \mathcal{M}'} = \text{LLH}_{\mathcal{M}} - \text{LLH}_{\mathcal{M}'}$$

- parameters of the model are unknown couplings $\mathcal{M}\{h\}$
- evaluation of the LLH for the optimized values $\mathcal{M}\{\hat{h}\}$

Example: the SM Higgs decay

Higgs decay to ZZ : $J^P = 0^+$

The term in the SM lagrangian:

$$\mathcal{L}_{ZZH} = \frac{m_Z^2}{2v} Z_\mu Z^\mu H$$

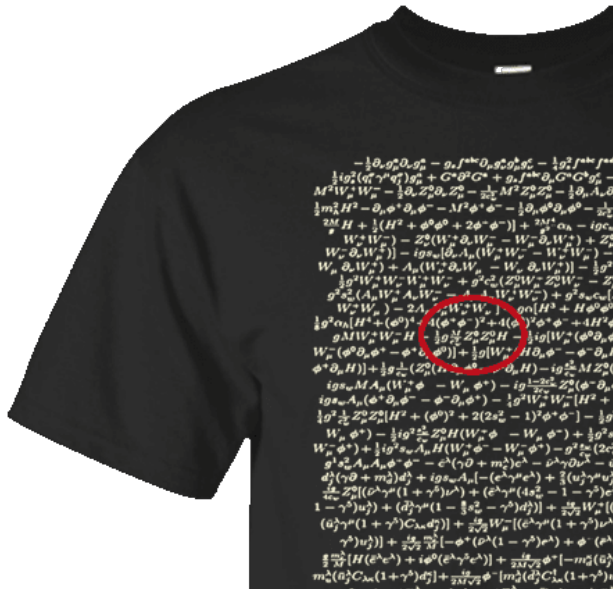
Helicity amplitude:

$$A_{\lambda_1, \lambda_2}^{H \rightarrow ZZ} = \frac{m_Z^2}{2v} (\epsilon_1^*(\lambda_1) \cdot \epsilon_2^*(\lambda_2)).$$

Matrix of couplings:

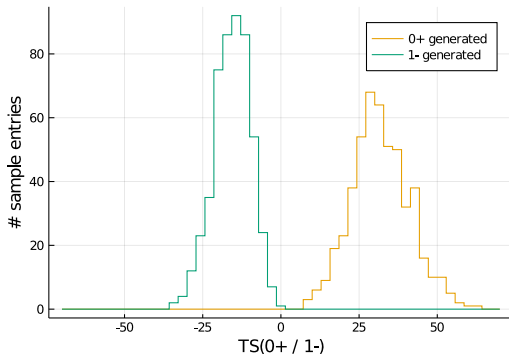
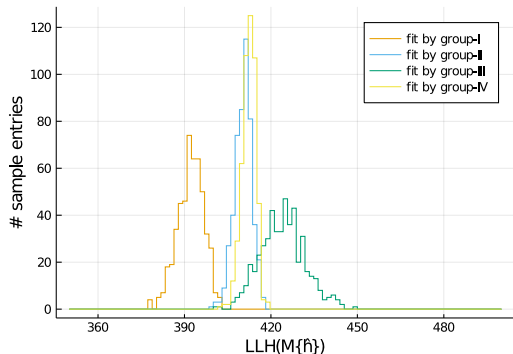
$$H_{\lambda_1, \lambda_2} \Rightarrow \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

First group: $b = d = 1$.



Numerical result: $H \rightarrow Z(\mu^+\mu^-)Z(\mu^+\mu^-)$

fits of a sample with 500 events. 500 MC experiments.



- The group-I has the highest likelihood (Yes!) $\Rightarrow J^P$ Natural, Even
- Width of distribution is due to the statistics
- LLH barely overlaps. TS separation is even better.

Discriminating observable: ϕ -dependence

How angular distributions are different

Net ϕ -dependence distinguishes naturality

$$\frac{2\pi}{N} \frac{dI}{d\phi} = 1 + \frac{h_{1,1} h_{-1,-1}^*}{2} \cos(2\phi),$$

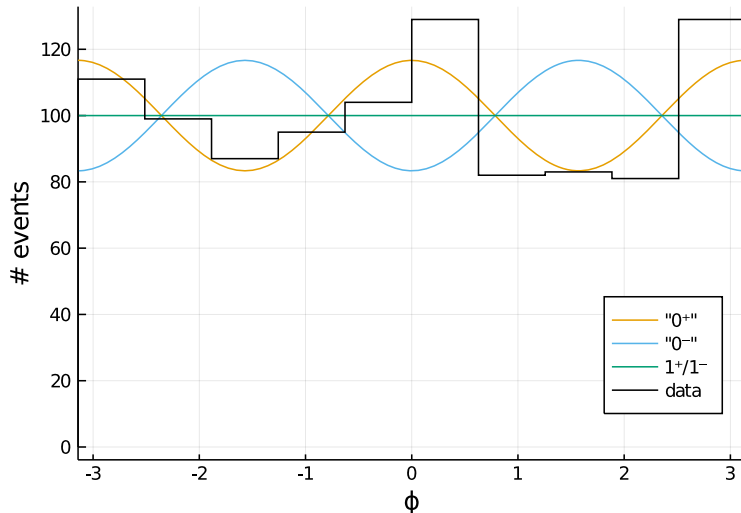
$$\sum_{\lambda_1, \lambda_2} |H_{\lambda_1 \lambda_2}|^2 = 1$$

groups	$h_{1,1} h_{-1,-1}^*$	$dN/d\phi (2\pi/N)$
group-I	$ b ^2$	$1 + b ^2 \cos(2\phi)/2$
group-II	$- b ^2$	$1 - b ^2 \cos(2\phi)/2$
group-III	0	flat
group-IV	0	flat

- either fit $dN/d\phi$ or calculate the moment $\langle \cos(2\phi) \rangle$
- distinguish groups I vs II vs (III & IV)
- warning: b might be zero $\Rightarrow$ works only if $\langle \cos(2\phi) \rangle \neq 0$

Higgs ϕ distribution

MS sample, 1000 events



Helicity matrix $H = \mathbb{I}/\sqrt{3}$,

$$h_{1,1}h_{-1,-1}^* = \frac{1}{3}$$

positive $\langle \cos(2\phi) \rangle$ moment,

$$\frac{2\pi}{N} \frac{dN}{d\phi} = 1 + \frac{1}{6} \cos(2\phi)$$

- Clearly different to the best one gets with other hypotheses

Summary

- Amplitude $X \rightarrow V(\mu\mu)V(\mu\mu)$ depends on 3 angles (5 angles) for the unpolarized (polarized) production.
- Decay of $X \rightarrow VV$ is fully determined by 3×3 matrix of the helicity couplings
- Parity + permutation symmetry constraint the matrix (+ particle-2 phase)

$$H_{\lambda_1, \lambda_2} = (-1)^J \eta_X H_{-\lambda_1, -\lambda_2},$$

$$H_{\lambda_1, \lambda_2} = (-1)^J H_{\lambda_2, \lambda_1}.$$

- There are four groups of quantum numbers, J^P : odd vs even, natural vs unnatural
- Likelihood test is the most powerful method
- ϕ -dependence is single-variable discriminating observable